

Gender Differences and Social Design in Human-AI Collaboration: Insights from Virtual Cobot Interactions Under Varying Task Loads

Carolin Wienrich
Psychology of Intelligent Interactive
Systems at University of Würzburg
Würzburg, Germany
carolin.wienrich@uni-wuerzburg.de

Marc Erich Latoschik
Human-Computer Interaction Group
at University of Würzburg
Würzburg, Germany
marc.latoschik@uni-wuerzburg.de

David Obremski
Psychology of Intelligent Interactive
Systems at University of Würzburg
Würzburg, Germany
david.obremski@uni-wuerzburg.de



Figure 1: Snapshots from the VR evaluation of Human-Robot Interaction based on the eXtended AI method. A: the cobot in its interaction area. B: user's interaction area with sorting bins and dispatch panel (enlarged for readability). C: cobot close-up.

ABSTRACT

This work explores the effects of users' gender and social design features of AI under different task load conditions on human-like attributions, social impact, work performance and perceived workload, user experience, and various other measures in Human-AI Interaction (HAI). Users had to execute sorting and dispatch tasks in collaboration with a virtual cobot. The degree of social gestalt of the cobot was varied by the ability to make small talk (i.e., talkative vs. non-talkative cobot), and the task load was increased by adding a secondary task (i.e., high vs. low task load condition). Overall, the talkative cobot led to a more positive perception of the cobot and increased social qualities like sense of meaning and team membership compared to the non-talkative cobot. The following gender effect was particularly interesting. The talkative cobot had a *buffering effect* for women and a *distraction conflict effect* for men in high task load conditions. When interacting with the talkative robot, women find the high task condition less stressful. In contrast thereto, the talkative cobot was distracting for men in the high task load condition. Our results highlight that social design choices and interindividual differences influence a successful collaboration

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650827>

between humans and AI. The work also shows the added value of systematic XR-simulations for the investigation and design of human-centered HAIs (eXtended AI approach).

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**;
Empirical studies in interaction design.

KEYWORDS

Human-AI Collaboration, Artificial Intelligence, Human Agent Interaction, Interindividual Differences, Gender, Collaborative Robots

ACM Reference Format:

Carolin Wienrich, Marc Erich Latoschik, and David Obremski. 2024. Gender Differences and Social Design in Human-AI Collaboration: Insights from Virtual Cobot Interactions Under Varying Task Loads. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3613905.3650827>

1 INTRODUCTION

Integrating robots and artificial interaction partners in modern work environments, a notion popularized by characters like WALL-E in cinema, has traditionally prioritized efficiency and performance [22]. Historically, the focus in human-AI interaction (i.e., HAI) has been on functionality, with less attention to the social dimensions and interindividual differences [37, 42]. However, as interactions with AI become more frequent, the importance of social aspects

and the recognition of individual differences, such as gender, are becoming increasingly vital in interaction design [12, 40]. Social design features can be operationalized by various cues, such as conversational cues [31]. There are initial indications that the ability to make small talk, for example, has a positive effect, especially for female users [46]. However, the design of additional features that go beyond pure functionality could also lead to distraction and, therefore, lower work performance, especially when the task load increases [3, 17, 24]. Even though this is an important design conflict, there is little research on how additional AI features, which might improve social AI design (e.g., conversational skills), impact performance in scenarios with high task demands. This paper aims to bridge this gap by investigating the impact of social design features and interindividual differences in collaborative robots (cobots) on HAI, particularly under varying levels of task load. To set the interaction in a realistic work setting and still be able to vary the cobot and task loads systematically, the interaction was simulated in virtual reality. Thus, the current work addresses the critical need to understand how these social design elements and individual differences influence HAI in practical settings.

2 RELATED WORK

Social aspects of HAI such as team spirit, appraisals, or fulfillment of needs, become increasingly important [9, 10]. This motivates the development of AIs that incorporate social intelligence [4] to some extent as a means to support the interactions with the users [31]. Initiating a conversation, adaptive answering, or sharing personal experiences are proven cues that imply the social intelligence of inanimate counterparts [1, 41] with positive effects on team perception [13, 18, 32], attribution of positive properties [8, 41], human performance, and usability [33]. Due to the positive social effects, this paper focuses on conversational skills as social design features of AI.

However, these effects have mostly been studied in isolation from other important aspects of HAI, e.g., different workload levels, which are most likely to occur in real-life scenarios. Therefore, investigating the impact of social AI features on perceived interaction in tasks of varying workload levels seems to be highly relevant, but has not yet been done.

Task load in the context of HAI is determined by several aspects [37]. Non-AI-specific aspects include the task to be worked on, e.g., the number of subtasks, the task complexity, or shortage of completion time [21, 34], as well as environmental conditions such as noise or lighting [15], or general ergonomic constraints.

AI-specific aspects of perceived workload in the context of HAI are determined by the general attitudes of the users towards digital tools in general and AI specifically, as well as the concrete designs of the human-AI interface that will be used for the task. Here, the emotional and social effects of HAI have a notable impact on performance and satisfaction [37, 47, 48]. While the general assumption is that these effects can be utilized for their positive influence (see above), social features, such as communication, contact, or proximity, might also be perceived as an additional load factor (see research on the distraction conflict theory like [3, 17, 24]). There is first evidence that besides positive effects on hedonic and social quality and perceived performance, there also is a negative

impact of conversational abilities of an AI on pragmatic quality and mental effort [46], which potentially could be amplified under high task load conditions.

However, a conclusive assessment of such impacts is still pending since other evidence shows that the inclusion of social features could also have a buffering effect under such conditions [11, 18, 39], which would then lead to positive influences. Overall, the design of social intelligence (i.e., conversational skills) in HAI has not yet been investigated in relation to various task load conditions. Thus, it remains unclear whether additional social design features have a distraction effect or a buffering effect, representing a gap in research.

It has also not been researched whether the same AI feature can have a distraction effect for some users and a buffering effect for others. Even though some recent studies show an influence of **interindividual differences** of users on the perception of AI [28, 44], they have not yet been investigated in connection with social design features under different workload conditions which represents a further gap in the research.

One of the factors contributing to the lack of research that addresses the huge design space of HAI and social aspects in combination with interindividual differences is the inherent difficulty in systematically modifying the behavior and appearance of artificial interaction counterparts and interaction paradigms in general.

Virtual, Mixed, and Augmented Reality (VR, MR, and AR: XR for short) has already been applied extensively in studying human-human interactions [5–7, 36, 45] or human-AI interactions [46]. Utilizing **XR as a testbed** for HAI enables the rapid prototyping of interactive and embodied AI entities (such as cobots). It allows for a systematic and credible examination of HAI variations in realistic contexts. Despite initial promising applications [46], the vast potential of altering the form of the AI system, the assigned task, or the environment concurrently has yet to be fully exploited.

2.1 Summary and Present Study

In summary, social aspects become increasingly important for perceived interaction effects and, hence, overall effective HAI. Initial studies suggest that social AI features positively impact perceived interaction. Other approaches postulate that additional design features can also have a distracting and, thus, negative effect on task performance, especially when the task load increases.

Thus, it seems plausible that the positive effects of social AI features can alleviate the perceived workload. Conversely, it is conceivable that a more social HAI may intensify the perceived workload. To systematically and validly investigate the opposing possibilities, we simulated a human-cobot interaction in VR and manipulated both social cobot features and task load. We also investigated how individual aspects influence the interplay between social cobot features and task load.

3 METHOD

3.1 Apparatus, Tasks, Factors

We designed an immersive VR exposure to explore the human-robot collaboration in a simulated sorting and dispatch task within an appropriate task environment. The VR environment was developed

with Unity Version 2019.2.13f1¹ for the HTC Vive Pro VR headset². The cobot was implemented using the gender-neutral version of *Ronin* model from the unity asset store³ (see Figure 1 A and C). In application, the model is roughly the size of a fully grown human. To enable the cobot to speak, 67 short audio files were created.

Each contains individual sentences and statements for creating and sustaining social presence (i.e. the feeling to be present in the communication) [1]. The experimenter played the audio files using the Wizard-of-Oz method [29] to achieve a flexible conversation flow. Hence, we distinguished cobot designs with two levels of social capabilities, one talkative cobot and one without verbal communication (factor **cobot**). This means that the experimenter played appropriate audio files according to the flow of the conversation, while the study participants assumed that the robot was speaking autonomously.

Examples are:

Starting Conversation

- Hi, I'm Roni. Looks like we're working together today.
- What's your name?
- ...

Helping Conversation

- Mmh, what letter does it have?
- This goes in the left-hand shaft to A-M.
- ...

Normal Conversation

- How long have you been working here? I don't think we've met yet.
- So, do you like working here?
- Unfortunately, there is no study permit for robots yet. Otherwise I would really study geology or archaeology.
- ...

Emergency sentences and Filler Words

- We can talk about it again later.
- Hm, unfortunately I don't know that either.
- Mmh, cool, agreed, ...
- ...

The VR application simulates an industrial hall: one side belongs to the cobot, including containers that are filled with colored packages and a digital display board on which the correct sequence of the packages is displayed. On the other side, two metal shafts ('A-M' and 'N-Z') are simulated, in which the packages are to be sorted by the participants, as well as another display board showing the number of correctly and incorrectly sorted packages (see Figure 1 A and B). These elements belong to the main task of sorting packages.

The additional display for the secondary task (i.e., monitoring the truck dispatch) is located to the left of the metal shafts (see Figure 1 enlarged area). The second task was chosen in such a way that it fits the content of the main task and meets the theoretical requirements of a second task (multiple resource theory of Wickens [43]). First of all, participants had to decide whether packages on the display should be sent to the north or south in dependency of

color (north: blue and turquoise; south: pink and yellow). Further, participants see two fuel gauges of two trucks. As soon as the tank of one of the two trucks is full and a red light is illuminated, they should give a signal as soon as possible.

In addition, participants were also asked to observe the number of packages that were loaded onto the trucks (northbound truck with 30 blue and 15 turquoise, and southbound truck with 20 yellow and 25 pink packages). As soon as the number of colored packages is reached, an acoustic signal is emitted, to which the participants should react as quickly as possible. Once the maximum number of packages per truck has been reached and the truck is completely filled, the participants should again give a signal that the truck has been dispatched and a new one can be loaded. To create additional time pressure, a five-minute countdown was displayed. We distinguished two task load conditions, one with the secondary task and one without (factor **task load**).

3.2 Measurement Materials

We assessed the impact of our experimental manipulations and interindividual differences (i.e., gender of users) on (I) attitudes and human-like attributions with respect to the cobot, (II) social impact, (III) work-related aspects and (IV) user experience. In addition, we checked for the success of the (V) experimental manipulations. Control questions were used to check for answering quality.

Additional information on the following questionnaires and items can be found in the supplemental material.

3.2.1 I. Attitudes and Human-like Attributions: (1) Attitude towards robots was assessed by the *Negative Attitude toward Robots Scale (NARS)* [35]. The original questionnaire consists of 14 items distributed across three subscales, which capture different dimensions of negative attitudes toward robots, i.e. *Negative Attitudes toward Situations and Interactions of Robots*, *Negative Attitudes toward Social Influence of Robots*, and *Negative Attitudes toward Emotions in Interaction with Robots*. Participants respond to the items on a 5-point Likert scale ranging from 'strongly disagree' to 'strongly agree'. We rephrase the items so they fit the virtual cobot and the items do not ask for a negative attitude per se but for a neutral one. A total score was calculated for data analyses and inverted, so that higher values indicate more positive attitudes. **(2) Human-like attitudes** assigned to the cobot were assessed by two questionnaires: On the one hand, we used the *Mind Attribution Scale (MAS)* to gauge various facets of mind attribution, such as intention, consciousness, emotions, and beliefs [30]. Participants responded on a 7-point Likert scale ranging from 'strongly disagree' to 'strongly agree'. A total score was calculated for data analyses. The higher the values, the more human-like attributions have been assigned. On the other hand, we used the *Uncanny Valley Scale* assessing respondents' feelings, attitudes, and perceptions towards human-like AIs [23]. The scale includes three dimensions: *human likeness*, *attractiveness*, and *eeriness*. Participants indicated their answers on a semantic differential. The semantic differential was converted into a 5-level scale for analyses. High values indicate more human likeness, attractiveness, or eeriness of the virtual cobot.

3.2.2 II. Social Impact: (1) Social presence was measured by three scales. The Bailenson scale [2] measured aspects of co-presence.

¹<https://unity.com/releases/editor/whats-new/2019.2.13>

²<https://www.vive.com/de/product/vive-pro-full-kit/>

³<https://assetstore.unity.com/packages/3d/characters/robots/cool-sci-fi-robot-drone-model-rigged-and-animated-124206>

Behavioral involvement was measured by the corresponding subscale of the game experience questionnaire [26]. In addition the sense of human contact was measured [14]. The 6-point Likert scale included five items, ranging from 'strongly disagree' to 'strongly agree'. (2) **Team spirit** perceived by the subjects was assessed by a questionnaire including seven subscales each containing two items [16]. However, we built an overall score by averaging all items for data analyses. Participants indicated their answers on a 5-point Likert scale, ranging from 'strongly disagree' to 'strongly agree'. (3) **Treatment after shift** was measured by asking subjects what they wanted to happen to the cobot after the shift. Pre-defined answers ranged from *The robot remains on the factory floor waiting for the next shift.* to *The robot is invited along for an after-work beer.* The participants indicated their level of agreement by choosing one of five statements which have been treated as 5-point Likert scale for data analyses. The higher the value the more human-like was the treatment. (4) **Social role** measurement asked subjects to specify which role the virtual cobot plays for them [38]. The social roles were *buddy, boyfriend or girlfriend, family member, colleague, companion, personal assistant, or tool.* The participants indicated their level of agreement on a 5-point Likert scale ranging from 'strongly disagree' to 'strongly agree'. We averaged the scale so that higher values indicate more human-like role assignment.

3.2.3 III. Work-Related Measurement: Count of the number of correctly sorted packages in addition to the perceived workload (described below) have been assessed as the objective **working performance** in the main task.

3.2.4 IV. User Experience: We used the short version of the *AttrakDiff* [20], including the pragmatic and hedonic aspect. They were extended by an eudaimonic aspect [25] and a social dimension [46]. They reflect the level of need fulfillment through the interaction with the virtual cobot. All items are positively polarized and are measured using a 7-point Likert scale. In addition to the four dimensions, a total score was calculated by averaging all items. The higher the score the more the need was met.

3.2.5 V. Manipulation Checks: (1) **Task load** (subjective) was measured by the *NASA Task Load Index (NASA-TLX)* [19]. We included all six dimensions: *Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration.* Each of the six subscales operates on a 100-point bipolar scale. Participants weighted the scores to compare different tasks or systems in terms of their workload. The unweighted value of a dimension is multiplied by its specific weight. In addition, we formed a total value of overall subscales. We used the weighted version for analyses. (2) **Level of biomorphism** was rated by the participants to check for the manipulation of the cobot design, i.e., of the conversational abilities of the cobot (item: *Please rate the following features of the cobot*). Subjects indicated their answers on a semantic differential, ranging from 'abstract' to 'biomorph'. The semantic differential was converted into a 10-level scale for analyses. High values indicate more biomorphism.

3.3 Participants

The participants included 54 students (female: 25, male: 29) from the University of Würzburg with an average age of 21.30 years

($SD = 1.90$). A power analysis (80% power, small effect size, significance level of .05) yielded a sample size of 52 participants. We recruited 60 participants from a local student portal and excluded six participants who failed the control questions or did not enter identical code words between the measurements. The final sample of 54 participants still met the requirements of our power analysis. Ethics approval was obtained before we began recruiting participants. Participants received course credit as compensation for participating in the hour-long study.

3.4 Procedure

The experiment was carried out in a standard office-like laboratory room with two computers, one for the VR and one for the questionnaire application.

As soon as participants registered for the study, they were sent a link to a questionnaire assessing sociodemographic data. In the laboratory, a test run with four packages was conducted before the experiment to practice the interaction in VR and ensure task understanding. In the test run, the cobot was not involved.

After the test run, participants interacted twice with the virtual cobot, once with cobot communication and once without. The order of the within-factor cobot was counterbalanced. Each interaction lasted approximately 5 minutes. Depending on the between-factor task load, randomly assigned, participants only had to either sort packages (low task load) or also operate the secondary task (high task load). After five minutes a bell rang. The participants could now take down the head-mounted display and evaluate the first experimental condition. Subsequently, the second, the same but opposite run took place, either with or without communication from the cobot. After each run, the numbers of correctly and incorrectly sorted packages were noted for later statistical analyses.

3.5 Data Analyses

The main interaction effects of the experimental conditions *cobot* and *task load* were analyzed with a 2x2 mixed ANOVA.

To analyze the impact of user gender, difference variables for the dependent variables were calculated. We built the differences between the conditions with the talkative cobot and the non-talkative cobot (within factor). Thus, positive values indicate higher ratings in the talkative cobot condition.

Then, a 2x2 between ANOVA with the factors *task load* and *gender* was conducted to analyze the main effect of participant gender and the interaction effect between task load and participant gender. The interaction effect of the difference variables shows whether the talkative cobot has more of a buffering or a disruptive function as the task load increases. Furthermore, it shows whether this depends on participant gender. The analyses were done with JASP [27].

4 RESULTS

4.1 Manipulation Checks

The communication style was rated significantly more biomorphic in the talkative cobot condition, $M = 7.88$, $SD = 2.44$, compared to the non-talkative cobot condition, $M = 3.16$, $SD = 2.32$, $F(1, 56) = 133.66$, $p < .001$, $\eta_p^2 = 0.71$. The overall perceived workload was significantly higher in the high task load condition, $M = 55.40$,

$SD = 11.72$, than in the low task load condition, $M = 19.44$, $SD = 12.00$, $F(1, 56) = 42.42$, $p < .001$, $\eta_p^2 = 0.43$. Consequently, the manipulation checks for cobot and task load manipulations were successful.

4.2 Impact of Social Cobot Design and Task Load

4.2.1 Attitudes and Human-like Attributions: In line with the expectations, the talkative cobot was rated more positively than the non-talkative one $F(1, 56) = 13.59$, $p < .001$, $\eta_p^2 = 0.20$. In contrast, the task load did not impact the **evaluation** of the virtual cobot, $F(1, 56) = 0.40$, $p = .53$, $\eta_p^2 = 0.01$.

As for social presence, the results show that subjects assigned more human-like characteristics to the talking cobot. The talkative cobot led to significantly more attribution of human mind $F(1, 56) = 64.59$, $p < .001$, $\eta_p^2 = 0.54$ humanness ($F(1, 56) = 66.24$, $p < .001$, $\eta_p^2 = 0.54$) and attractiveness ($F(1, 56) = 6.64$, $p = .013$, $\eta_p^2 = 0.11$), but also to more eeriness ($F(1, 56) = 86.40$, $p < .001$, $\eta_p^2 = 0.61$) than the non-talkative one (see Figure 2).

In contrast, the **attribution of human-like characteristics** was mainly not affected by the task load conditions (mind attribution: $F(1, 56) = 0.16$, $p = .70$, $\eta_p^2 < 0.01$; subscale attractiveness: $F(1, 56) = 0.29$, $p = .586$, $\eta_p^2 < 0.01$; subscale eeriness: $F(1, 56) = 1.16$, $p = .186$, $\eta_p^2 = 0.02$). There is one exception: the higher task load led to a higher attribution of humanness $F(1, 56) = 5.54$, $p = .02$, $\eta_p^2 = 0.09$.

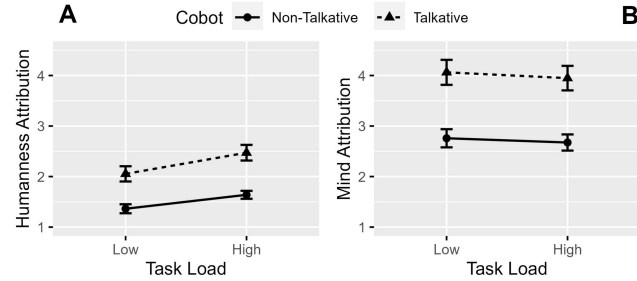


Figure 2: Effects of cobot design and task load on (A) humanness and (B) mind attribution.

4.2.2 Social Impact: As expected, the **social presence** was significantly higher for the talkative cobot than for the non-talkative cobot in all three sub measures (co-presence: $F(1, 56) = 92.72$, $p < .001$, $\eta_p^2 = 0.62$; behavioral involvement: $F(1, 56) = 78.56$, $p < .001$, $\eta_p^2 = 0.58$; sense of human contact: $F(1, 56) = 160.31$, $p < .001$, $\eta_p^2 = 0.74$). The task load condition showed less impact on **social presence**. While task load did not impact co-presence ($F(1, 56) = .28$, $p = .60$, $\eta_p^2 < 0.01$) or sense of human contact ($F(1, 56) = .01$, $p = .92$, $\eta_p^2 > 0.01$), the high task load condition led to less behavioral involvement than the low task condition ($F(1, 56) = 6.63$, $p = .001$, $\eta_p^2 = 0.12$).

As expected, the talkative cobot significantly led to higher **team spirit** $F(1, 56) = 27.84$, $p < .001$, $\eta_p^2 = 0.55$ compared to the non-talkative one (see Figure 3). Similarly, the **treatment after**

shift was significantly more positive and human-like (joint beer after the shift) for the talkative cobot than for the non-talkative cobot (storage with other machines), $F(1, 56) = 20.79$, $p < .001$, $\eta_p^2 = 0.27$. In contrast, the task load did not impact **team spirit** ($F(1, 56) = 0.04$, $p = .85$, $\eta_p^2 = 0.05$) or **treatment after shift** ($F(1, 56) = 3.08$, $p = .085$, $\eta_p^2 = 0.05$).

Finally, participants attributed more social roles like *workmate* and fewer machine-like roles like *tool* to the talkative cobot compared to the non-talkative cobot $F(1, 56) = 83.32$, $p < .001$, $\eta_p^2 = 0.60$. Similarly, the higher task load led to more **assignment of human-like roles** $F(1, 56) = 8.78$, $p = .027$, $\eta_p^2 = 0.08$.

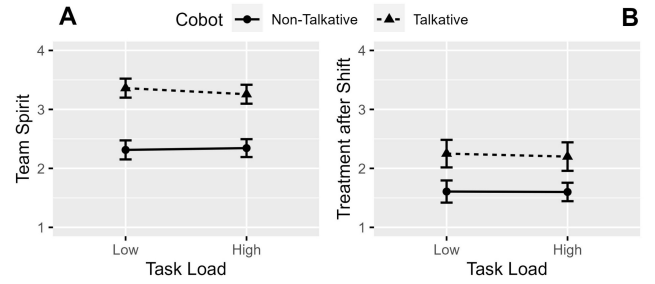


Figure 3: Effects of cobot design and task load on (A) team spirit and (B) treatment after shift.

4.2.3 Work-related Measures: Overall, the social cobot design did not significantly affect the **perceived workload**. However, one interesting effect was observable. The talkative cobot significantly led to a lower perceived mental workload $F(1, 56) = 157.41$, $p < .001$, $\eta_p^2 = 0.74$ than the non-talkative one (see Figure 4). In contrast, high task load led to more **perceived workload** as described in the manipulation check session. The significant increase of perceived workload in the high task condition was observable in the following NASA-TLX subscales: mental workload: $F(1, 56) = 53.58$, $p < .001$, $\eta_p^2 = 0.49$; temporal workload: $F(1, 56) = 47.85$, $p < .001$, $\eta_p^2 = 0.46$; performance: $F(1, 56) = 39.35$, $p < .001$, $\eta_p^2 = 0.41$; effort: $F(1, 56) = 57.77$, $p < .001$, $\eta_p^2 = 0.51$ and frustration: $F(1, 56) = 5.15$, $p = .027$, $\eta_p^2 = 0.08$.

The social cobot design did not significantly affect the **working performance**. However, and as expected, participants sorted fewer packages correctly in the high task load condition than in the low task load condition, $F(1, 56) = 109.75$, $p < .001$, $\eta_p^2 = 0.66$.

4.2.4 User Experience: As expected, the talkative cobot significantly led to higher social need $F(1, 56) = 30.89$, $p < .001$, $\eta_p^2 = 0.35$ and hedonic need $F(1, 56) = 3.94$, $p = .05$, $\eta_p^2 = 0.06$ fulfillment compared to the non-talkative one. However, the social cobot design did not significantly affect pragmatic, and eudaimonic UX aspects (see Figure 4). Surprisingly, the task load did not impact the fulfillment of pragmatic, hedonic, or social needs. However, higher task load led to higher eudaimonic need fulfillment than lower task load $F(1, 56) = 4.32$, $p = .04$, $\eta_p^2 = 0.07$.

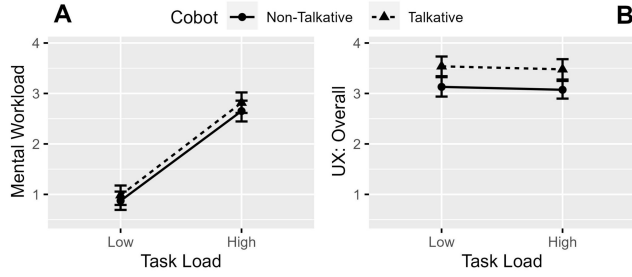


Figure 4: Effects of cobot design and task load on (A) mental workload and (B) UX.

4.3 Interaction between Social Cobot Design and Task Load Conditions

Overall, the social cobot design did not significantly buffer or increase the **perceived workload** in different task load conditions, $F(1, 56) = .12, p = .734, \eta_p^2 < 0.01$. However, one interesting effect was observable. The talkative cobot significantly increased the perceived workload, particularly in the high task load condition $F(1, 56) = 4.39, p = .04, \eta_p^2 = 0.07$. There were no other significant interaction effects.

4.4 Interindividual Characteristics: Gender

4.4.1 Attitudes and Human-like Attributions: Regarding the evaluation of the virtual cobot, women and men rated the cobot similarly over all conditions. However, simple main effects showed that the talkative cobot has a buffering effect for women compared to men in the high task load condition regarding the **evaluation of the cobot** (low task load: $F(1, 54) = 0.04, p = .851$; high task load: $F(1) = 2.48, p = .032$). While the evaluation became more negative in the high task load condition with the talkative cobot for men, women's perception was getting more positive (see Figure 5). Other variables describing the **attribution of human-like characteristics** or the **assignment of human-like roles** were not affected by participant gender.

4.4.2 Social Impact: Compared to men, women reported more **social presence** in the talkative cobot condition. The effect was only significant in the co-presence scale ($F(1, 54) = 5.65, p = .02, \eta_p^2 = 0.10$).

However, simple main effects revealed that the talkative cobot has a buffering effect for women compared to men in the high task load condition regarding behavioral involvement (low task load: $F(1, 54) = 3.34, p = .073$; high task load: $F(1, 54) = 10.44, p = .002$). While men reported less behavioral involvement in the high task load condition with the talkative cobot, women reported more behavioral involvement (see Figure 5).

Similar to co-presence, women reported more **team experience** in the communication condition compared to men $F(1, 54) = 6.91, p = .04, \eta_p^2 = 0.08$. The **treatment after shift** was not significantly affected by participant gender.

4.4.3 Work-related Measures: Participant gender did not impact the **perceived workload**. However, simple main effects showed that the talkative cobot has a buffering effect for women compared

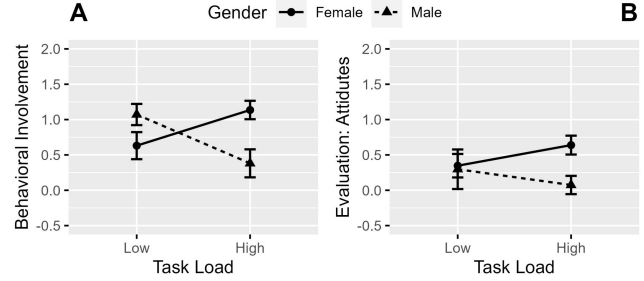


Figure 5: Effects of social cobot by task load and participant gender on (A) social presence and (B) evaluation.

to men in the high task load condition regarding the **working performance** (low task load: $F(1, 54) = 1.64, p = .206$; high task load: $F(1) = 4.38, p = .041$). While men sorted fewer packages correctly in the high task load condition with the talkative cobot, women sorted more packages correctly (see Figure 6).

4.4.4 User Experience: Overall, women tended to report more user experience in the talkative cobot condition than men. The gender effect was significant for eudaimonic need fulfillment, $F(1, 54) = 6.91, p = .01, \eta_p^2 = 0.11$. In addition, simple main effects showed that the talkative cobot has a buffering effect for women compared to men in the high task load condition regarding the pragmatic (low task load: $F(1, 54) = 0.202, p = .655$; high task load: $F(1) = 3.99, p = .051$) and the eudaimonic need fulfillment (low task load: $F(1, 54) = 3.06, p = .082$; high task load: $F(1) = 3.90, p = .053$). While the need fulfillment decreased with a higher task load for men, it increased for women in the talkative cobot condition (see Figure 6).

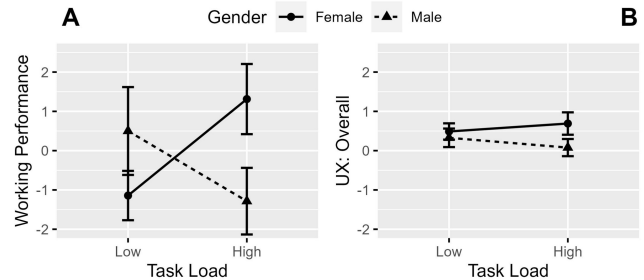


Figure 6: Effects of social cobot by task load and participant gender on (A) performance and (B) UX.

5 DISCUSSION, LIMITATIONS, AND CONCLUSION

The results reveal numerous positive effects of the talkative cobot. Thus, the social cobot design features led to a more positive evaluation, a better team spirit, and a better UX. These results align with previous studies reporting positive effects of social design features [8, 13, 18, 32, 33, 41].

While the general assumption is that these effects can be utilized for their positive influence (see above), social features, such as

communication, contact, or proximity, might also be perceived as an additional stress factor (see research on the distraction conflict theory like [3, 17, 24]).

Initially, neither a *buffering effect* [37, 47, 48] nor distraction conflict [3, 17, 24] could be detected in the interaction between social cobot design features and a higher task load. The only apparent interaction effect was that the task appeared to be more effortful with the talkative cobot in the high task condition. However, when participant gender is taken into account, very different patterns emerge. Women generally respond more positively to the talkative cobot by reporting a higher social presence, user experience, and fulfillment of eudaimonic needs. It also shows that the talkative cobot particularly supports women when the task load increases (more behavioral involvement, higher pragmatic and eudaimonic need fulfillment). They even perform objectively better at the task. These results, therefore, indicate a *buffering effect* for women. In contrast, men rate the talkative cobot lower when the task load increases. Surprisingly, it even has a negative effect on their objective work performance. Men are, therefore, more likely to exhibit a *distraction conflict*.

A limitation of the current study is that the aspects of the talkative cobot that contributed to the *buffering* or *distraction effect* can not be further specified. It remains unclear whether this was due to the language interaction, the topics, or if it was related to the task. Furthermore, it should be investigated if the effects of social design features persist over a longer interaction time. In future studies, how social design features could be implemented should be explored so that they also have a positive effect on men. In addition, it should be investigated whether other interindividual aspects, such as cultural background or personality traits also influence the results. Since the present study was limited to having university students as participants, a larger and broader sample should be examined for this purpose. Finally, the transferability of the results to a real rather than a simulated interaction should be examined. Nevertheless, it was shown that the human-centered approach of simulating human-AI interactions in XR (eXtended AI approach [46]) and thus investigating influencing factors systematically and controllable in relatively realistic contexts is promising and can enrich future research in the field.

In conclusion, the current work clearly shows that not only efficacy and pragmatism are important in the design of human-AI interactions, but that social design features and interindividual differences are important influencing factors. The systematic variation of social design characteristics under different task load conditions contributes, in particular, to a better understanding of human AI collaborations and the inclusion of individual needs in the design of future workplaces. Our results clearly show that gender differences between users should be considered in the design of AI.

ACKNOWLEDGMENTS

This research has been funded by the German Ministry of Science and Education (BMBF) in the project HiAvA (Grant 16SV8781).

REFERENCES

- [1] Steven R Aragon. 2003. Creating social presence in online environments. *New directions for adult and continuing education* 2003, 100 (2003), 57–68.
- [2] Jeremy N Bailenson, Eyal Aharoni, Andrew C Beall, Rosanna E Guadagno, Aleksandar Dimov, and Jim Blascovich. 2004. Comparing behavioral and self-report measures of embodied agents' social presence in immersive virtual environments. In *Proceedings of the 7th Annual International Workshop on PRESENCE*, Vol. 1105. IEEE.
- [3] Robert S Baron. 1986. Distraction-conflict theory: Progress and problems. *Advances in experimental social psychology* 19 (1986), 1–40.
- [4] Frank Biocca, Chad Harms, and Judee K Burgoon. 2003. Toward a more robust theory and measure of social presence: Review and suggested criteria. *Presence: Teleoperators & virtual environments* 12, 5 (2003), 456–480.
- [5] Jim Blascovich. 2002. Social influence within immersive virtual environments. In *The social life of avatars: Presence and interaction in shared virtual environments*. Springer, 127–145.
- [6] Jim Blascovich, Jack Loomis, Andrew C Beall, Kimberly R Swinth, Crystal L Hoyt, and Jeremy N Bailenson. 2002. Immersive virtual environment technology as a methodological tool for social psychology. *Psychological inquiry* 13, 2 (2002), 103–124.
- [7] Dario Bombari, Marianne Schmid Mast, Elena Canadas, and Manuel Bachmann. 2015. Studying social interactions through immersive virtual environment technology: virtues, pitfalls, and future challenges. *Frontiers in psychology* 6 (2015), 869.
- [8] Cynthia Breazeal. 2004. Social interactions in HRI: the robot view. *IEEE transactions on systems, man, and cybernetics, part C (applications and reviews)* 34, 2 (2004), 181–186.
- [9] Cynthia Breazeal, Kerstin Dautenhahn, and Takayuki Kanda. 2016. Social robotics. *Springer handbook of robotics* (2016), 1935–1972.
- [10] Astrid Carolus and Carolin Wienrich. 2021. Towards a holistic approach and measurement of humans interacting with speech-based technology. In *1st AI-debate workshop: Workshop establishing an interdisciplinary perspective on speech-based technology*. 39–41.
- [11] Julie Carpenter. 2013. Just doesn't look right: Exploring the impact of humanoid robot integration into explosive ordnance disposal teams. In *Handbook of research on technoself: Identity in a technological society*. IGI Global, 609–636.
- [12] Jie Chen, Yulong Ding, Bin Xin, Qingkai Yang, and Hao Fang. 2020. A unifying framework for human-agent collaborative systems—Part I: Element and relation analysis. *IEEE Transactions on Cybernetics* 52, 1 (2020), 138–151.
- [13] Jessie YC Chen, Shan G Lakhmani, Kimberly Stowers, Anthony R Selkowitz, Julia L Wright, and Michael Barnes. 2018. Situation awareness-based agent transparency and human-autonomy teaming effectiveness. *Theoretical issues in ergonomics science* 19, 3 (2018), 259–282.
- [14] Dianne Cyr, Khaled Hassanein, Milena Head, and Alex Ivanov. 2007. The role of social presence in establishing loyalty in e-service environments. *Interacting with computers* 19, 1 (2007), 43–56.
- [15] Mathias Diebig, Franziska Jungmann, Andreas Müller, and Ines Catharina Wulf. 2018. Inhalts- und prozessbezogene Anforderungen an die Gefährdungsbeurteilung psychischer Belastung im Kontext Industrie 4.0. *Zeitschrift für Arbeits- und Organisationspsychologie A&O* (2018).
- [16] Josef A Fischer and Hendrik Hüttermann. 2020. Speak-Up-Check (SUC): Fragebogen zur Messung von proaktiver und konstruktiver Kommunikation in Teams. In *Zusammenstellung sozialwissenschaftlicher Items und Skalen (ZIS)*.
- [17] Bernard Guerin and JM Innes. 1984. Explanations of social facilitation: A review. *Current psychological research & reviews* 3 (1984), 32–52.
- [18] Svyatoslav Guznov, J Lyons, Marc Pfahler, A Heironimus, Montana Woolley, Jeremy Friedman, and A Neimeier. 2020. Robot transparency and team orientation effects on human-robot teaming. *International Journal of Human-Computer Interaction* 36, 7 (2020), 650–660.
- [19] Sandra G Hart and Lowell E Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology*. Vol. 52. Elsevier, 139–183.
- [20] Marc Hassenzahl and Andrew Monk. 2010. The inference of perceived usability from beauty. *Human-Computer Interaction* 25, 3 (2010), 235–260.
- [21] Keith C Hendy, Jianqiao Liao, and Paul Milgram. 1997. Combining time and intensity effects in assessing operator information-processing load. *Human Factors* 39, 1 (1997), 30–47.
- [22] Clint Heyer. 2010. Human-robot interaction and future industrial robotics applications. In *2010 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 4749–4754.
- [23] Chin-Chang Ho and Karl F MacDorman. 2010. Revisiting the uncanny valley theory: Developing and validating an alternative to the Godspeed indices. *Computers in Human Behavior* 26, 6 (2010), 1508–1518.
- [24] Pascal Huguet, Marie P Galvaing, Jean M Monteil, and Florence Dumas. 1999. Social presence effects in the Stroop task: further evidence for an attentional view of social facilitation. *Journal of personality and social psychology* 77, 5 (1999), 1011.
- [25] Veronika Huta. 2016. An overview of hedonic and eudaimonic well-being concepts. *The Routledge handbook of media use and well-being* (2016), 14–33.
- [26] WA IJsselstein, Y De Kort, and K Poels. 2013. The Game Experience Questionnaire. Technische Universiteit Eindhoven. *Eindhoven* (2013), 3–9.
- [27] JASP Team. 2024. JASP (Version 0.18.3)[Computer software]. <https://jasp-stats.org/>

- [28] Alexandra D Kaplan, Theresa T Kessler, J Christopher Brill, and PA Hancock. 2023. Trust in artificial intelligence: Meta-analytic findings. *Human factors* 65, 2 (2023), 337–359.
- [29] John F Kelley. 1983. An empirical methodology for writing user-friendly natural language computer applications. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. 193–196.
- [30] Megan N Kozak, Abigail A Marsh, and Daniel M Wegner. 2006. What do I think you're doing? Action identification and mind attribution. *Journal of personality and social psychology* 90, 4 (2006), 543.
- [31] Mengjun Li and Ayoung Suh. 2022. Anthropomorphism in AI-enabled technology: A literature review. *Electronic Markets* 32, 4 (2022), 2245–2275.
- [32] Joseph B Lyons, Sean Mahoney, Kevin T Wynne, and Mark A Roebke. 2018. Viewing machines as teammates: A qualitative study. In *2018 AAAI Spring Symposium Series*.
- [33] Joseph E Mercado, Michael A Rupp, Jessie YC Chen, Michael J Barnes, Daniel Barber, and Katelyn Procci. 2016. Intelligent agent transparency in human-agent teaming for Multi-UxV management. *Human factors* 58, 3 (2016), 401–415.
- [34] Kathleen L Mosier, Nikita Sethi, Shane McCauley, Len Khoo, and Judith M Orasanu. 2007. What you don't know can hurt you: Factors impacting diagnosis in the automated cockpit. *Human Factors* 49, 2 (2007), 300–310.
- [35] Tatsuya Nomura, Tomohiro Suzuki, Takayuki Kanda, and Kensuke Kato. 2006. Measurement of negative attitudes toward robots. *Interaction Studies. Social Behaviour and Communication in Biological and Artificial Systems* 7, 3 (2006), 437–454.
- [36] Xueni Pan and Antonia F de C Hamilton. 2018. Why and how to use virtual reality to study human social interaction: The challenges of exploring a new research landscape. *British Journal of Psychology* 109, 3 (2018), 395–417.
- [37] Matthew S Prewett, Ryan C Johnson, Kristin N Saboe, Linda R Elliott, and Michael D Covert. 2010. Managing workload in human-robot interaction: A review of empirical studies. *Computers in Human Behavior* 26, 5 (2010), 840–856.
- [38] Amanda Purington, Jessie G Taft, Shruti Sannon, Natalya N Bazarova, and Samuel Hardman Taylor. 2017. "Alexa is my new BFF" social roles, user satisfaction, and personification of the Amazon Echo. In *Proceedings of the 2017 CHI conference extended abstracts on human factors in computing systems*. 2853–2859.
- [39] Lionel P Robert Jr and Sangseok You. 2015. Subgroup formation in teams working with robots. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems*. 2097–2102.
- [40] Samantha Straka, Martin Jakobus Koch, Astrid Carolus, Marc Erich Latoschik, and Carolin Wienrich. 2023. How do employees imagine AI they want to work with: A drawing study. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–8.
- [41] Cristen Torrey, Aaron Powers, Matthew Marge, Susan R Fussell, and Sara Kiesler. 2006. Effects of adaptive robot dialogue on information exchange and social relations. In *Proceedings of the 1st ACM SIGCHI/SIGART conference on Human-robot interaction*. 126–133.
- [42] Valeria Villani, Fabio Pini, Francesco Leali, and Cristian Secchi. 2018. Survey on human-robot collaboration in industrial settings: Safety, intuitive interfaces and applications. *Mechatronics* 55 (2018), 248–266.
- [43] Christopher D Wickens. 2002. Multiple resources and performance prediction. *Theoretical issues in ergonomics science* 3, 2 (2002), 159–177.
- [44] Carolin Wienrich, Astrid Carolus, David Roth-Isigkeit, and Andreas Hotho. 2022. Inhibitors and Enablers to Explainable AI Success: A Systematic Examination of Explanation Complexity and Individual Characteristics. *Multimodal Technologies and Interaction* 6, 12 (2022), 106.
- [45] Carolin Wienrich, Nina Döllinger, and Rebecca Hein. 2021. Behavioral framework of immersive technologies (BehaveFIT): How and why virtual reality can support behavioral change processes. *Frontiers in Virtual Reality* 2 (2021), 627194.
- [46] Carolin Wienrich and Marc Erich Latoschik. 2021. extended artificial intelligence: New prospects of human-ai interaction research. *Frontiers in Virtual Reality* 2 (2021), 686783.
- [47] Beverly Woolf, Winslow Burleson, Ivon Arroyo, Toby Dragon, David Cooper, and Rosalind Picard. 2009. Affect-aware tutors: recognising and responding to student affect. *International Journal of Learning Technology* 4, 3-4 (2009), 129–164.
- [48] Euijung Yang and Michael C Dorneich. 2017. The emotional, cognitive, physiological, and performance effects of variable time delay in robotic teleoperation. *International Journal of Social Robotics* 9 (2017), 491–508.